

(a)



Base LLM model

1

Fine-tuning with Conditional Multi-Task Loss

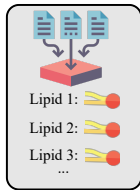
2

Prompt: "You are an experienced lipid expert. Please help me predict:xxx"



Predictor Agent

(b)



Virtual Library

①



Predictor Agent

None Toxic,
Efficient Score is 8.
Reason is <structure a>

②



No



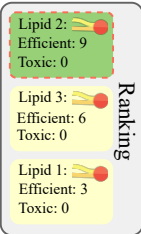
③



No

Yes

High efficient
lipid we want!



Predicted Library

④

Yes

Wrong.
Reason is
<structure b>

Let me check reason
and score...



Verifier Agent

⑤

⑥

No

If loop > 3

Yes

Let me decide ...

⑦



Human